SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: INTERNET PROGRAMMING
COURSE CODE: CSA4304

COURSE OUTCOME COVERED

CO-1: Develop well-structured, standards-compliant web pages using HTML/XHTML and XML, and apply Internet protocols and HTTP communication mechanisms for web content delivery. **(L2)**

Assessment Tool:Code along exercise: Make your first HTML page together with CSS

Weightage: 10%

QUESTIONS: **Total Marks: 10**

Scenario: Web Development Standards Implementation

A start up is developing a modern web application. During development, the team encounters multiple issues related to HTML elements, HTTP methods, XML syntax, and web communication protocols.
The following observations are made:

- The team needs to embed a promotional video on the homepage.
- A registration form must securely send user data to the server.
- The navigation menu must follow semantic HTML standards.
- XML configuration files are being used but contain syntax errors.
- The application must communicate over the correct web protocol.

Data Snippets Provided

Snippet 1: Video Embedding Options

A. <media src="video.mp4"></media>
B. <video src="video.mp4" controls></video>
C. <movie src="video.mp4"></movie>
D. <vid src="video.mp4"></vid>

Snippet 2: Form Submission Methods

A. GET
B. POST
C. PUT
D. DELETE

Snippet 3: Navigation Elements

A.
B. <nav>
C. <section>
D. <header>

Snippet 4: XML Syntax

A. <note><to>User</to><from>Admin</from>
B. <note><to>User</to><from>Admin</from></note>
C. <note><to>User</from></note>
D. <note to="User"></note

Snippet 5: Web Communication Protocols

A. FTP
B. HTTP
C. SMTP
D. SNMP

Questions:

1. Select the **correct HTML tag for video embedding** and justify your answer.
2. Identify the **appropriate HTTP method for form submission** and explain why.
3. Choose the **proper semantic element for navigation** and justify its usage.
4. Identify the **correct XML syntax** and explain the error in other options.
5. Choose the **correct protocol for web communication** and justify your choice.

1. Select the correct HTML tag for video embedding and justify your answer.

Correct Option: B

<video src="video.mp4" controls></video>

Explanation:

The <video> tag is the standard HTML5 element used to embed videos directly into a web page. The controls attribute provides built-in controls such as play, pause, volume, and fullscreen, allowing users to interact with the video easily. It is supported by all modern web browsers and does not require additional plugins.

Justification:

The other options (<media>, <movie>, and <vid>) are not valid HTML elements. Therefore, the <video> tag is the correct and standards-compliant choice for embedding a promotional video.

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2. **Identify the appropriate HTTP method for form submission and explain why.**

Correct Option: B. POST

Explanation:
The **POST** method is used to send data from a client to a server. It is the preferred method for registration forms because user information, such as names, email addresses, and passwords, is sent in the request body instead of the URL. This makes it more secure and suitable for handling sensitive data.

Justification:

- **GET** is mainly used to retrieve data and displays data in the URL.
- **PUT** is used to update existing resources.
- **DELETE** is used to remove resources from the server.

Hence, **POST** is the most appropriate method for user registration.

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3. Choose the proper semantic element for navigation and justify its usage.

Correct Option: B

<nav>

Explanation:
The <nav> element is a semantic HTML5 tag specifically designed to contain navigation links, such as menus and navigation bars. It helps browsers, search engines, and screen readers identify the navigation section of a webpage, improving accessibility and SEO.

Justification:
The element has no semantic meaning, while <section> and <header> serve different purposes. Therefore, <nav> is the correct element for creating a navigation menu.

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4. Identify the correct XML syntax and explain the errors in the other options.

Correct Option: B

```
<note>
  <to>User</to>
  <from>Admin</from>
</note>
```

Explanation:
This XML document is well-formed because it has a single root element, all opening tags have matching closing tags, and the elements are properly nested. Well-formed XML is essential for correct parsing and data exchange.

Errors in the other options:

- **Option A:** Missing the closing </note> tag.
- **Option C:** Incorrect closing tag (</from> instead of </to>).
- **Option D:** The closing tag is incomplete because the final > is missing.

Therefore, **Option B** is the only valid XML syntax.

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5. Choose the correct protocol for web communication and justify your choice.

Correct Option: B. HTTP

Explanation:
HTTP (HyperText Transfer Protocol) is the standard protocol used for communication between web browsers and web servers. It enables the transfer of web pages, images, videos, and other web resources over the Internet. Most modern websites use **HTTPS**, which is the secure version of HTTP.

Justification:

- **FTP** is used for file transfer.
- **SMTP** is used for sending emails.
- **SNMP** is used for network management.

Therefore, **HTTP** is the correct protocol for web communication in a web application.

Code of Conduct:

*I **N.SAI PRANAV/192411183**(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: **SAI PRANAV**

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding ; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, wellstructured , and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand